



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df019-2017-ios
Operator:	Floris
Analysis Started:	2026-06-18 12:05:01
Analysis Completed:	2026-06-18 12:07:01
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2017\dfdof_output\ft-df019-2017-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df019_external.001	drone_sd	physical	14.8 GB	2026-06-18 12:05:58
ios_logical.zip	controller_ios	logical	232.0 MB	2026-06-18 12:05:58

Evidence Source Path

C:\Users\Floris\Documents\ddji_drones\test_mavic_pro\df019\2017\ios

Cryptographic Hashes

df019_external.001		drone_sd	physical
SHA-256:	01e18f7b1662e229077fac9b505f6e6a4e766762ed18f543c73e7076f4020053		
SHA-1:	5dc7a5d072c68e533fc02018be140552f6e7539a		

ios_logical.zip		controller_ios	logical
SHA-256:	0c5493518e337f87533e957a0e3212d51dcfd4f698bf7da2c935a5e161db9055		
SHA-1:	989aa1be6ac8879c09ac23e11c8ac95dced67985		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	14	11	[~] Partial
fls	1	1	[OK]
icat	6	6	[OK]
mmls	1	1	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	3	3	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 3
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	08RDD9100100KG	[K] Controller-only	4
Drone Model	Mavic Pro	[K] Controller-only	3
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	1
DJI Application Version	4.1.5	[K] Controller-only	4
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	13	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-08-26T17:49:56Z	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 8c8a62f8299111b0f7f16665c95c1be0129837766b74c03973feb5cbd08526bd
Flight Record Name: DJIFlightRecord_2017-08-29__11-49-33_.csv
Start: 2017-08-29T17:49:33.091000+00:00
End: 2017-08-29T17:50:13.533000+00:00
Duration: 40.4 s

Flight Metrics

Peak Height: 5.7 m
Distance Travelled: 1.79 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 48.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 17:49:33	ctrl_ios:fr (v4.1.5)	Log started	SN: 08RDD9100100KG; App: 4.1.5; Mavic Pro
2017-08-29 17:49:33	ctrl_ios:fr (v4.1.5)	Motor turned on	on
2017-08-29 17:49:33	ctrl_ios:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 17:49:46	ctrl_ios:fr (v4.1.5)	Reached peak height	5.7 m
2017-08-29 17:50:13	ctrl_ios:fr (v4.1.5)	Log ended	Dist home: 0.3 m; Height: -0.3 m; Rec: 48.0 s

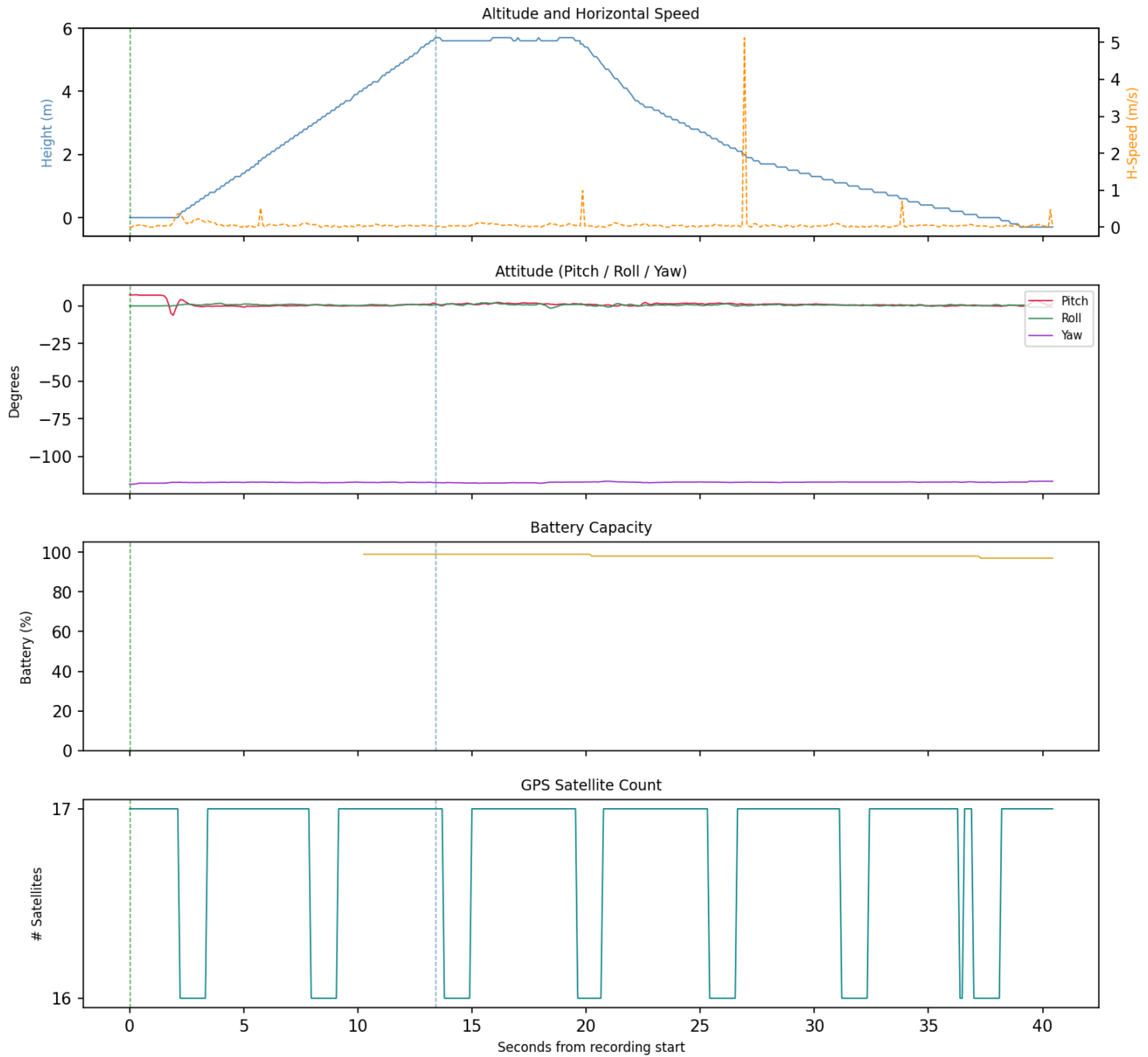
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:abec1888c705a197db50b92ec9cbaec7e754eccf113fbbeaf750f46e6837b8e8	

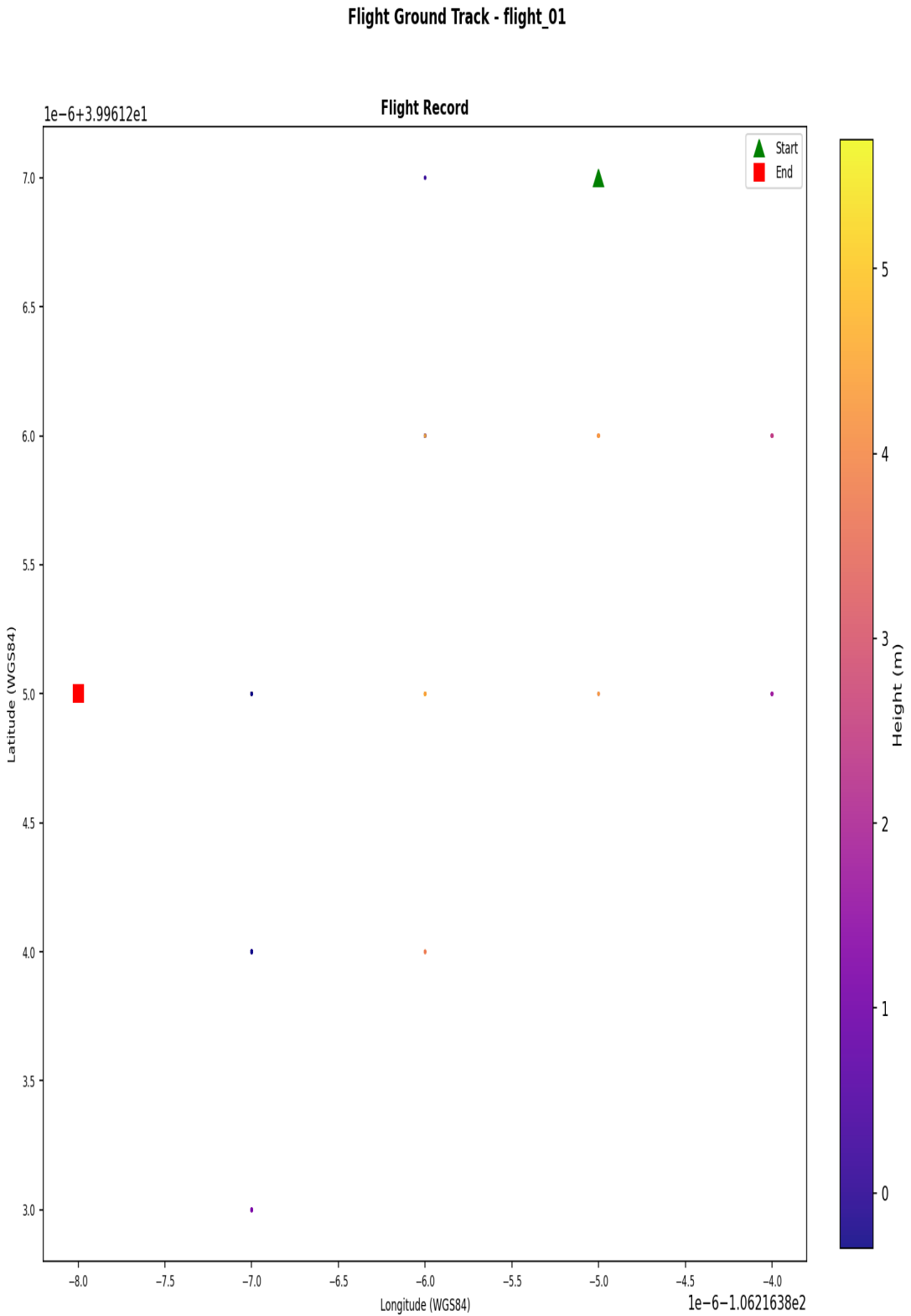
Flight Record - Sensor Telemetry

Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: a1184a01598e32feec6b8eff3105decc191a2ae76c85a88a4df568b3eb457765
Flight Record Name: DJIFlightRecord_2017-08-29__11-50-40_.csv
Start: 2017-08-29T17:50:40.861000+00:00
End: 2017-08-29T17:54:53.260000+00:00
Duration: 252.4 s

Flight Metrics

Peak Height: 29.6 m
Distance Travelled: 109.51 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 327.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. GoHome
5. AutoLanding
6. Joystick
7. NaviSubMode_Pointing
8. ForceLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 17:50:40	ctrl_ios:fr (v4.1.5)	Log started	SN: 08RDD9100100KG; App: 4.1.5; Mavic Pro
2017-08-29 17:50:40	ctrl_ios:fr (v4.1.5)	Motor turned on	on
2017-08-29 17:50:41	ctrl_ios:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 17:53:18	ctrl_ios:fr (v4.1.5)	Reached peak height	29.6 m
2017-08-29 17:54:53	ctrl_ios:fr (v4.1.5)	Log ended	Dist home: 0.2 m; Height: -0.1 m; Rec: 327.0 s

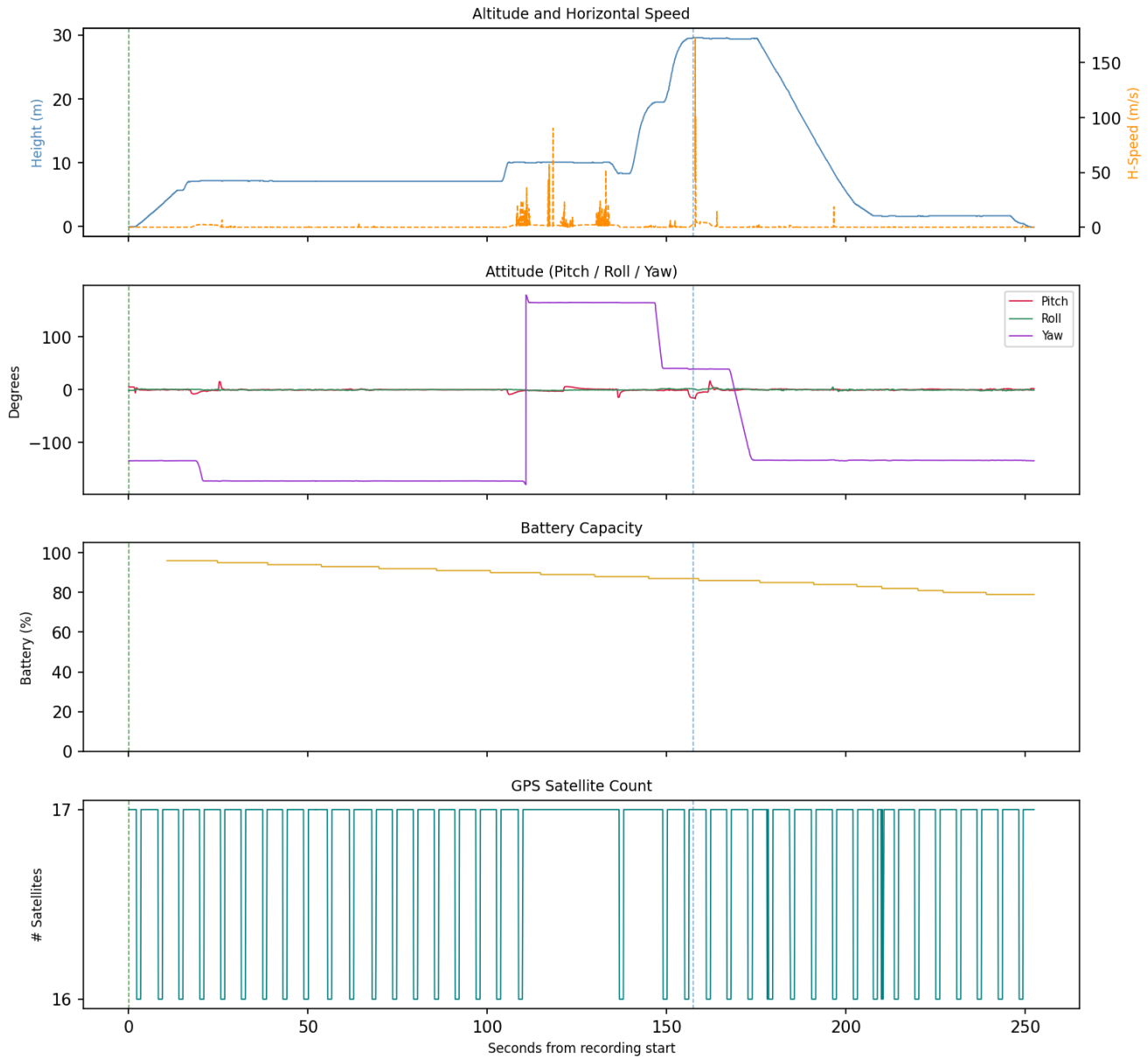
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:abec1888c705a197db50b92ec9cbaec7e754eccf113fbbef750f46e6837b8e8	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:b5dd962b7e4a556a2cf4f6ca1ebefbcbdafba5e041ee2d225e9ca5089bcedb7cf	

Flight Record - Sensor Telemetry

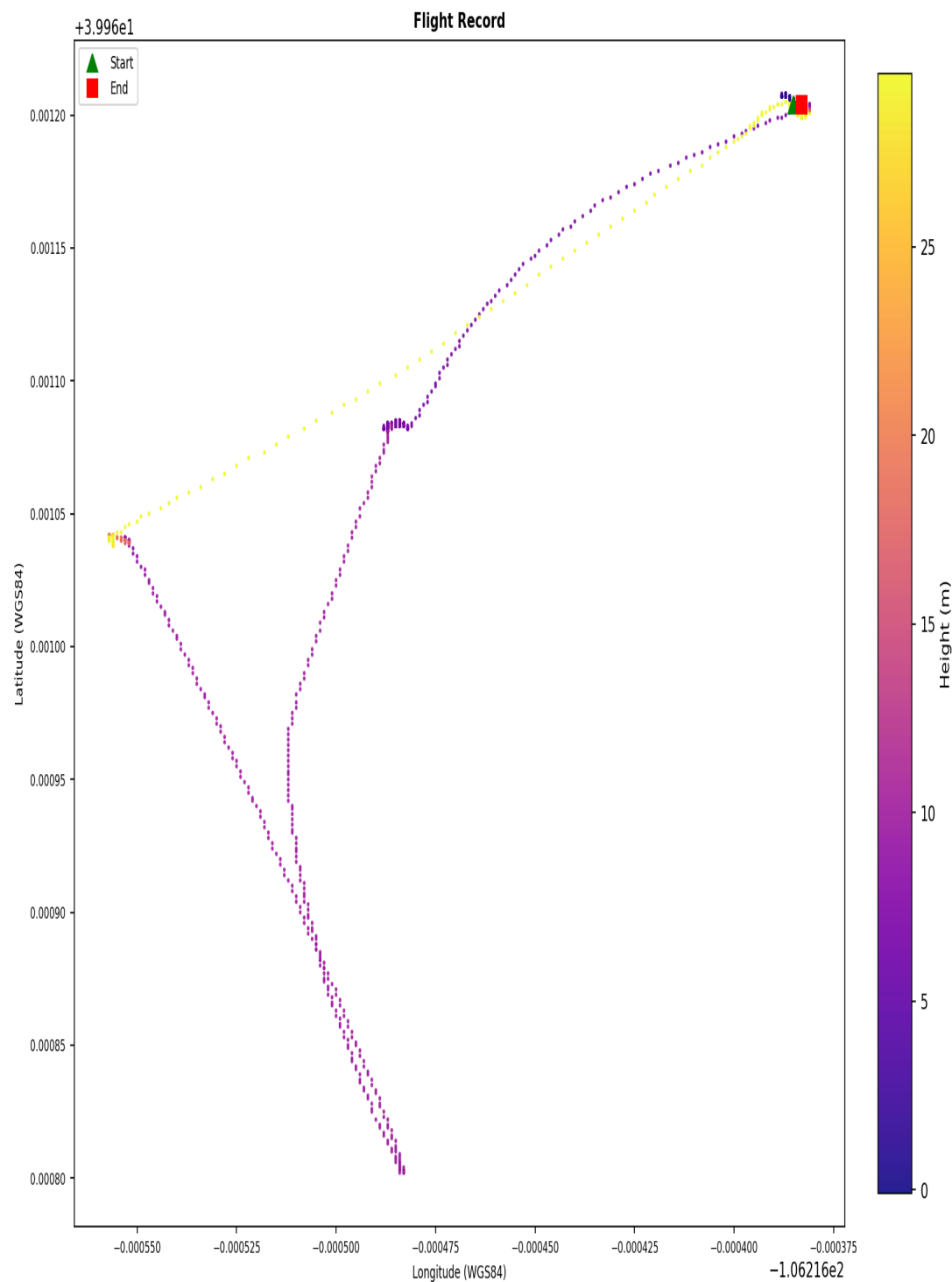
Flight Record Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_02



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 8eabd8c92e3a5b8f98990f073ec6b96ae3c529c31a776a4d771f334a787df359
Flight Record Name: DJIFlightRecord_2017-08-29__11-56-11_.csv
Start: 2017-08-29T17:56:11.305000+00:00
End: 2017-08-29T18:02:08.320000+00:00
Duration: 357.0 s

Flight Metrics

Peak Height: 15.3 m
Distance Travelled: 216.44 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 354.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. NaviSubMode_Pointing
5. AutoLanding
6. ForceLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 17:56:11	ctrl_ios:fr (v4.1.5)	Log started	SN: 08RDD9100100KG; App: 4.1.5; Mavic Pro
2017-08-29 17:56:11	ctrl_ios:fr (v4.1.5)	Motor turned on	on
2017-08-29 17:56:12	ctrl_ios:fr (v4.1.5)	Record mode changed	No
2017-08-29 17:56:14	ctrl_ios:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 17:58:53	ctrl_ios:fr (v4.1.5)	Reached peak height	15.3 m
2017-08-29 18:02:08	ctrl_ios:fr (v4.1.5)	Log ended	Dist home: 0.8 m; Height: 0.4 m; Rec: 354.0 s

In-Flight Warnings

Timestamp (UTC)	Source	Warning Message
2017-08-29 17:59:07	ctrl_ios:fr (v4.1.5)	Warning:Max Flight Distance Reached. Adjust in Main Controller Settings if necessary.
2017-08-29 17:59:35	ctrl_ios:fr (v4.1.5)	Warning:Max Flight Distance Reached. Adjust in Main Controller Settings if necessary.

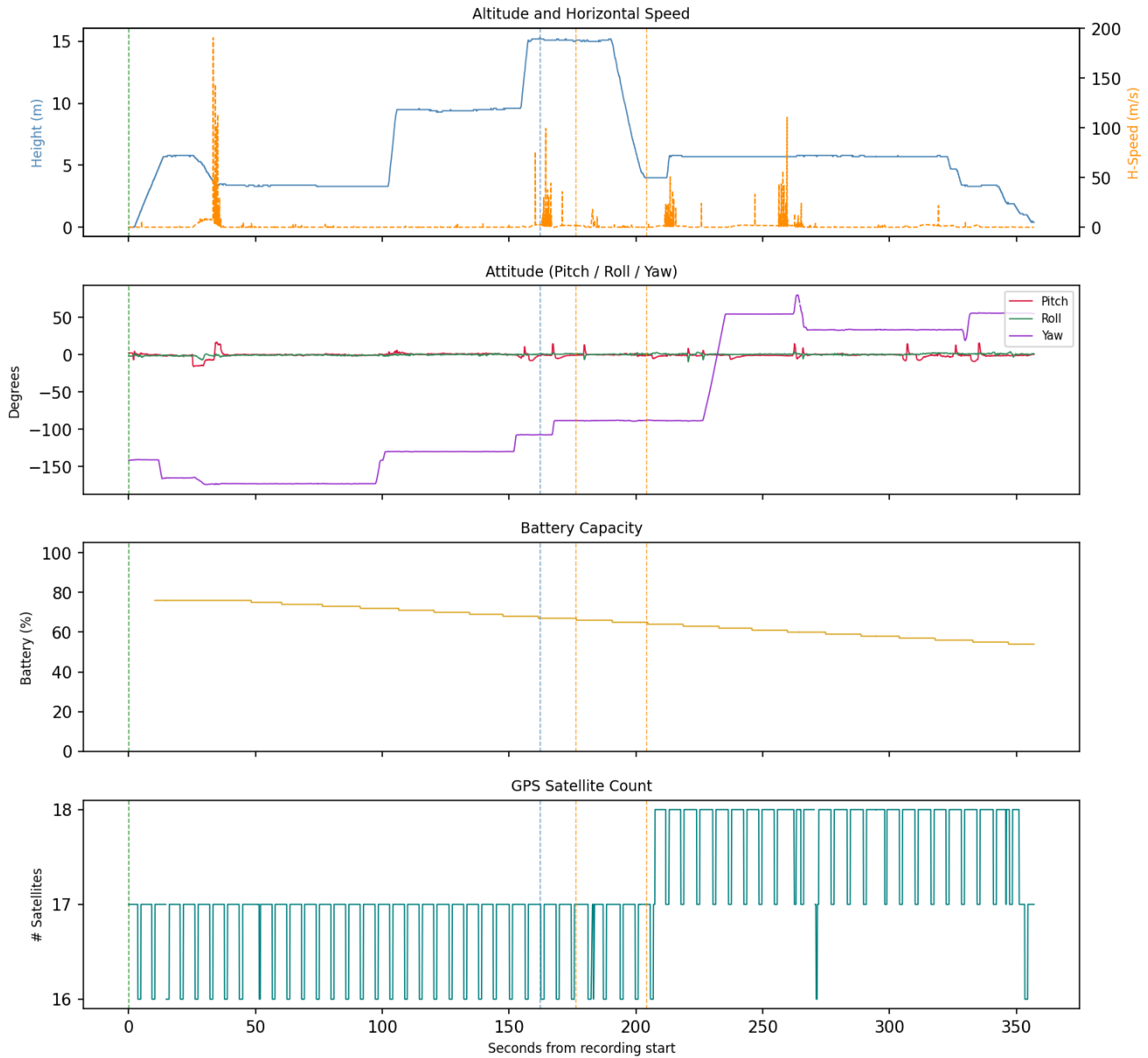
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:b5dd962b7e4a556a2cf4f6ca1ebefbcbdafba5e041ee2d225e9ca5089bcedb7cf	

Flight Record - Sensor Telemetry

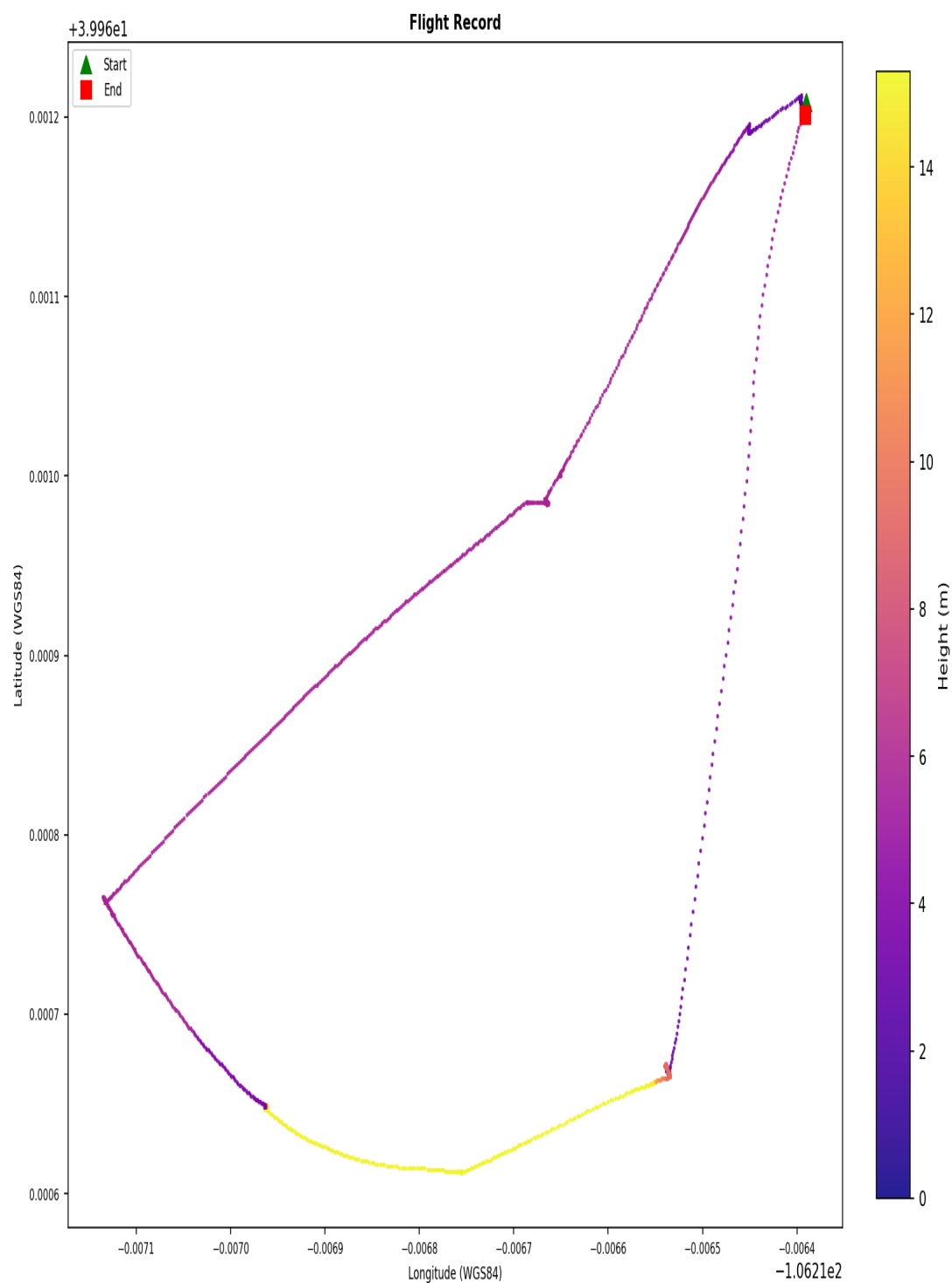
Flight Record Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_03



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	1	6	0	4	3	4	4
drone_sd	0	0	0	0	0	3	3

Data Quality Notes per Category

Databases (6 files):

- 2 database(s) empty (controller_ios)

Flight Logs (4 files):

- 1 file(s) unreadable (binary)
- formats: 1 bracketed_log, 2 json_array_log

Flight Records (3 files):

- 3 file(s) with row anomalies
- 3 file(s) with empty columns
- 3 file(s) with constant-value columns

Images (7 files):

- 7 file(s) missing (normalisable) EXIF timestamp
- 7 file(s) missing EXIF GPS coordinates

Videos (7 files):

- 3 file(s) contain(s) EXIF GPS coordinates
- 7 file(s) missing (normalisable) EXIF timestamp
- 4 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 bracketed_log, 2 json_array_log.
- Flight_records: 3 file(s) with row anomalies detected in telemetry data.
- Flight_records: 3 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 3 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

statistics.db

- database_empty: all tables empty

Flight Logs

2017-08-29_11_49_33-08RDD9100100KG.dat

- format: json_array_log | entries: 15

2017-08-29_11_56_11-08RDD9100100KG.dat

- format: json_array_log | entries: 13

upgradeLog_0.txt

- format: bracketed_log | entries: 192

Flight Records

norm_DJIFlightRecord_2017-08-29_11-49-33.csv

- altitude_negative: 20 row(s) affected
- altitude_spike: 2 row(s) affected
- contains_no_value: 57 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+52 more)
- contains_constant_value: 163 column(s) - APP_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+158 more)

norm_DJIFlightRecord_2017-08-29_11-50-40.csv

- altitude_spike: 8 row(s) affected
- contains_no_value: 57 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+52 more)
- contains_constant_value: 155 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType, CAMERA_INFO.firmUpgradeState (+150 more)

norm_DJIFlightRecord_2017-08-29_11-56-11.csv

- altitude_spike: 13 row(s) affected
- contains_no_value: 56 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+51 more)
- contains_constant_value: 154 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState,

CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+149 more)

Images**2017_08_29_11_49_25.thumbnail**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_56_13.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_56_59.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_59_15.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos**2017_08_29_11_49_25.mp4**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_56_13.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_56_59.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_11_59_15.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0002.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no firmware version found - Device and account information.
2. Note: no last known latitude (find aircraft) found - Device and account information.
3. Note: no last known longitude (find aircraft) found - Device and account information.
4. Note: no last known altitude (find aircraft) found - Device and account information.
5. Note: no last known heading (find aircraft) found - Device and account information.
6. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
7. 3 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03). Independent corroboration is absent for these flight(s).
8. 4 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	b010ed7f654a4496678722d4e15b07cc74f4892b381c363f6c9eacf2dee1c7be			
SHA-1:	a429b3d8cc03b4d8666c4c53d57032518b7483ba			

airmap_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c			
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	c3635c4495369db4324667ff800fa21938ab44d19853d7cf5c462d5795a4eebe			
SHA-1:	5fbc7f7d19f443b48975f45ee1a0ef6abb3f12f3			

nfz_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	3af048ec88d8e3523fd0a8ef775d708348913ed6ae5ed1feeb93a3d31325334d			
SHA-1:	1e1c834784b088adb20de1761785556ab9f0eede			

statistics.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdfa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

tencent_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	dc5faceb3d5509d4475d80d8559a7bfeFed36f1ef1dd30d191701f16568c8e90			
SHA-1:	204131865d963c85bf6c935234cacc19bb4b9c90			

tencent_wxo_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	dc88684c6e1dfc5c7cd07bdefa740e5afffae91cfc1e54bd8cdd2b47984024c9			
SHA-1:	0da793f6a282760fda369c482d09f68e98ccf907			

2017-08-29_11_49_33-08RDD9100100KG.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	80d4fb7edfc1815768bbdde8764b1c71e60def03f163691023a34f3a55007790			
SHA-1:	87166f604719ed5264bc95c21fc43755cd77bdf2			

2017-08-29_11_56_11-08RDD9100100KG.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	831ae86486717274464b0e3d32adffe253d7cb2ae7990e99a248f0af9fc8c146			
SHA-1:	32dfd0c5d5eb0b0c628a00b31ea6f98c04379016			

app_log.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	52bf2e2d07cc4e182dc7fd54ffa59393f3273ba94a8885d30466f8731d9d820d			
SHA-1:	eff9df436c8485e3e193a6e2bfac861dde986d27			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	7886db0f159aa6e661b308636b94bdb01a555659cb361bf308599b4a1e49abc			
SHA-1:	a0ac599d9d094db0cac9d2b568518c66c4e23015			

DJIFlightRecord_2017-08-29_11-49-33_.txt		Flight Records	controller_ios	extract_logical
SHA-256:	1e6abf634721c823a363a917e06df41c493d812f98a4c89dfef31bd3cc256c22			
SHA-1:	2a111cf8538a4bc0d0df9821c0c6c4048fc8863f			

DJIFlightRecord_2017-08-29_11-50-40_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	a722cffa25823257ad652d2a8e5134923c093897b498c6fef907ee483f1e8937		
SHA-1:	77889e2b4a36db65b4fb690f429e1d51de510c6e		

DJIFlightRecord_2017-08-29_11-56-11_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	6fb961ac585d91c2daf93500d8a850e4a1006fa56c8185b5fe659ad330d72f80		
SHA-1:	5314147f5c4082939c3ba3f226c0dce26c327753		

2017_08_29_11_49_25.thumbnail	Images	controller_ios	extract_logical
SHA-256:	a81a737dfbeea19beaa9af889674a0c6acc43dee962d07331344fe209d1c3327		
SHA-1:	461bce7110fd7e82574cca032daf05bbd8aef5e0		

2017_08_29_11_56_13.thumbnail	Images	controller_ios	extract_logical
SHA-256:	c4bf182b90aa0c4a369c87f4c0f02abd4a097b94ed76d737f00cd242abe6f46f		
SHA-1:	bbcf52c2a0d747d2970ac40fa43f34d5067f988f		

2017_08_29_11_56_59.thumbnail	Images	controller_ios	extract_logical
SHA-256:	ea755f7e7620315f9ec7690ad956ecbc30f63cedf606fa20f30ccefafa244b292f		
SHA-1:	c7c0dae3ba9065b57a0a7ac33c5f8348fb6e57be		

2017_08_29_11_59_15.thumbnail	Images	controller_ios	extract_logical
SHA-256:	122a474f027fda26fc77a79e78aacf474f94ef4713970ab7ff0f1bfff79ad7edd		
SHA-1:	3e6c4b78b66498293714aaa6275f3ef3fcf41e95		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	d6efab989c124c66c0e0549052fde15a1f72a3fcaafd26d4576f0002924bfb6e		
SHA-1:	5aec150e3cbda88b63275f6647db783d02cd683		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	e0e95d5acc8cbcd1bbe4dfca9a3d8a2a900f201c4185b6bba6f71aef0ae09a35		
SHA-1:	5624ae8614d28a6fdb2b3df43ba8a5a9df538133		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	a94693918341e4193895608acb10fa875e19d5d21be4c83b906c1acf41fdafa61		
SHA-1:	0a49299fc05680e62fd94bb3bd4e74fada9454b0		

2017_08_29_11_49_25.mp4	Videos	controller_ios	extract_logical
SHA-256:	a43a8d8b7b063df166c64748008f4657f896edcb08cbef452908f38700864a3a		
SHA-1:	300743cefed4ad74e851a90478ecba42bae10c3b		

2017_08_29_11_56_13.mp4	Videos	controller_ios	extract_logical
SHA-256:	0f7c8f62e8f8f01a07f8395fcfe62812387a2897028e2b76d4f6f973de8ccf1a		
SHA-1:	0d56289c003915acac6112f360a9976a3b562b55		

2017_08_29_11_56_59.mp4	Videos	controller_ios	extract_logical
SHA-256:	fd28f97f7758984e9c7e564128e96aa4aa9084d6d1e3133efae19e1d1eb9b42f		
SHA-1:	1cae870b84ac052fa503e8be6975b33e904bd162		

2017_08_29_11_59_15.mp4	Videos	controller_ios	extract_logical
SHA-256:	85ddc3a332f537cd2a10041f01ba06f11a14b252577dc2516a33a779cd9d90b5		
SHA-1:	1a9bd7165081b14c198851285711788119edcb6f		

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	abec1888c705a197db50b92ec9cbaec7e754eccf113fbbef750f46e6837b8e8			
SHA-1:	55100e0a109f3bbe44faa75b7c17bd18c4b890f8			

DJI_0002.MOV		Videos	drone_sd	extract_physical
SHA-256:	b5dd962b7e4a556a2cf4f6ca1ebefbcdafba5e041ee2d225e9ca5089bcdcb7cf			
SHA-1:	19f6485705dde70bcb0c23f280246fecb275d197			

DJI_0003.MOV		Videos	drone_sd	extract_physical
SHA-256:	29a5d9fd9021d15bcf7aaa8b0345474359f7e033d9917544b6b70c605ecce8db			
SHA-1:	7bb709881d6be660b056d8abe0f98887381f85c6			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	6	0	0
Flight Logs	4	0	0
Flight Records	3	3	3
Images	4	4	4
Videos	4	4	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	3	3	0
Videos	3	3	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 12:05:01 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df019_external.001

2026-06-18 12:05:13 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df019_external.001

2026-06-18 12:06:13 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df019-2017-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-18 12:06:21 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 8197
out: DJI_0001.MOV

2026-06-18 12:06:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 8198
out: DJI_0002.MOV

2026-06-18 12:06:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 8199
out: DJI_0003.MOV

2026-06-18 12:06:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 9222
out: DJI_0001.THM

2026-06-18 12:06:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 9223
out: DJI_0002.THM

2026-06-18 12:06:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df019_external.001 9224
out: DJI_0003.THM

2026-06-18 12:06:53 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-08-29__11-49-33.txt DJIFlightRecord_2017-08-29__11-49-33.csv
out: DJIFlightRecord_2017-08-29__11-49-33.csv

2026-06-18 12:06:53 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-08-29__11-50-40.txt DJIFlightRecord_2017-08-29__11-50-40.csv
out: DJIFlightRecord_2017-08-29__11-50-40.csv

2026-06-18 12:06:54 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-08-29__11-56-11.txt DJIFlightRecord_2017-08-29__11-56-11.csv
out: DJIFlightRecord_2017-08-29__11-56-11.csv

2026-06-18 12:06:55 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_49_25.thumbnail
out: 2017_08_29_11_49_25_exif.json

2026-06-18 12:06:55 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_56_13.thumbnail
out: 2017_08_29_11_56_13_exif.json

2026-06-18 12:06:55 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_56_59.thumbnail
out: 2017_08_29_11_56_59_exif.json

2026-06-18 12:06:56 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_59_15.thumbnail
out: 2017_08_29_11_59_15_exif.json

2026-06-18 12:06:56 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_49_25.mp4
out: 2017_08_29_11_49_25_exif.json

2026-06-18 12:06:56 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_56_13.mp4
out: 2017_08_29_11_56_13_exif.json

2026-06-18 12:06:57 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_56_59.mp4
out: 2017_08_29_11_56_59_exif.json

2026-06-18 12:06:57 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_11_59_15.mp4
out: 2017_08_29_11_59_15_exif.json

2026-06-18 12:06:57 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-18 12:06:58 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM
out: DJI_0002_exif.json

2026-06-18 12:06:58 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-18 12:06:58 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV
out: DJI_0001_exif.json

2026-06-18 12:06:59 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MOV
out: DJI_0002_exif.json

2026-06-18 12:06:59 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV
out: DJI_0003_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.